

Education

Carnegie Mellon University

MASTER'S IN ROBOTICS

- Incoming MSR student

Pittsburgh, PA, USA

Starting Sep. 2027

SJTU (Shanghai Jiao Tong University)

B.E. IN COMPUTER SCIENCE AND TECHNOLOGY

- IEEE Honors Class
- GPA: 90.6/100 (3.93/4.3)
- Relevant Coursework: Mathematics Analysis (90/100) · Linear Algebra (94/100) · Probability and Statistics (94/100) · Linear and Convex Optimization (96/100) · Information Theory (93/100) · Discrete Mathematics (91/100) · Robotics (96/100) · Operation System (95/100)

Shanghai, China

Sep. 2022 - Present

Publications

Tac-Bench: Benchmarking Diverse Tactile Manipulation and Multimodal World Models

PEILIN WU, ZHIYI LI, WEIHAN XU, HAONAN CHEN, YILUN DU

Jul. 2025 - Oct. 2025

RSS Workshop (Tactile Sensing for Robotic Foundation Models), 2026

- A controlled comparison of multimodal policy architectures, isolating fusion strategy from 59 confounding task and data choices, together with the imitation-learning baselines.
- A physical-realism protocol for multimodal world models that directly measures the physical utility through closed-loop rollout success and optimization-based reachability gains.

Structured 4D Latent Predictive Model for Robot Planning

ZHIYI LI, PEILIN WU, XIAOSHEN HAN, YILUN DU

Jul. 2025 - Oct. 2025

ICML, 2026

- A 4D latent world model that predicts future 3D structures conditioned on current observations and text goal, achieving high visual quality, physical consistency, and robust viewpoint generalization.
- A planning framework that leverages our model's detailed 3D predictions as geometrically rich goals for an inverse dynamics controller, enabling precise and spatially aware manipulation.

LoopSR: Looping Sim-and-Real for Lifelong Policy Adaptation of Legged Robots

PEILIN WU, WEIJI XIE, JIAHANG CAO, HANG LAI, WEINAN ZHANG

Jun. 2024 - Mar. 2025

IROS, 2025

- Modeled the lifelong learning problem and proposed methods accordingly to get a simulated reconstruction of the real world.
- Designed a lifelong policy adaptation framework that enhanced the performance by at least 30% in the most difficult cases compared with sim-to-real transfer baselines and successfully handled problems like catastrophic forgetting.

Bridging the Sim-to-Real Gap from the Information Bottleneck Perspective

HAORAN HE, PEILIN WU, CHENJIA BAI, HANG LAI, LINGXIAO WANG, LING PAN, XIAOLIN HU, WEINAN ZHANG

Apr. - Jun. 2024

CoRL (Oral), 2024

- Provided a theoretical analysis to model the sim-to-real gap concerning privileged information and historical trajectories.
- Proposed an efficient and effective sim-to-real transfer method inspired by information bottleneck, outperforming existing baselines (DreamWaq, RMA, etc.) for about 10% in simulated RL tasks and real-world quadruped locomotion.

Research Experience

Harvard University

RESEARCH INTERN AT HARVARD, ADVISED BY PROF. YILUN DU

Boston, U.S.

Jul. 2025 - Present

Research Topic: Diffusion-based World Models, Multimodal Perception, 3D Representation

Shanghai AI Lab

RESEARCH ASSISTANT AT EMBODIED AI CENTER, ADVISED BY DR. JINGBO WANG

Shanghai, China

Mar. 2025 - Present

Research Topic: Human Motion Generation, Humanoid Whole Body Control

Shanghai Jiao Tong University

RESEARCH ASSISTANT AT APEX LAB, ADVISED BY PROF. WEINAN ZHANG

Shanghai, China

Sep. 2023 - Present

Research Topic: Reinforcement Learning, Legged Robots, Continual Learning

Shanghai Jiao Tong University

RESEARCH ASSISTANT AT MAGIC LAB, ADVISED BY PROF. SIHENG CHEN

Shanghai, China

Jul. 2023 - Mar. 2024

Research Topic: Drone System, Collaborative Communication

Highlighted Projects

World Model for Robot Manipulation with Tactile Information

Boston, U.S.

RESEARCH ASSISTANT AT EMBODIED AI CENTER, ADVISED BY DR. JINGBO WANG

July. 2025 - Present

- Construct a multimodal world model to generate both visual and tactile information through diffusion-based techniques.
- Develop policies for contact-rich robot manipulation tasks under the supervision of the world model.

Bridging the Gap between Human Motion Generation and Humanoid Control

Shanghai, China

RESEARCH ASSISTANT AT EMBODIED AI CENTER, ADVISED BY DR. JINGBO WANG

Mar. 2025 - Present

- Established a thorough pipeline from text/goal-conditioned motion generation to low-level locomotion of humanoid robots (Unitree G1).
- Implementing RL-based fine-tuning techniques to get a robust system that can continuously improve.

Drone System Construct and Communication for UAV swarm

Shanghai, China

RESEARCH ASSISTANT AT MAGIC LAB, ADVISED BY PROF. SIHENG CHEN

Aug. 2023 - Mar. 2024

- Constructed a drone system based on ROS, carrying a GPS sensor and USB camera, used to collect data for autonomous driving datasets.
- Implemented the communication for UAV swarm based on TCP/UDP, preparing for future research on collaborative communication.

Vision System for 6-DoF Robot Arm

Shenzhen, China

TEAM MEMBER AT SJTU ROBOMASTER TEAM IN ROBOMASTER COMPETITION 2023

Feb. - Aug. 2023

- Designed the algorithm to identify the camera pose and desired end effector pose from RGB image.
- Constructed a thorough pipeline to connect the sensor, PC, and lower computer based on ROS and serial communication.

Honors & Awards

2023 **1st Prize**, 1st in 32 teams, China University Robot Competition RoboMaster

2022-2024 **Zhiyuan Honorary Scholarship**, top 5 % students in SJTU

2023, 2024 **University Scholarship**, top 5 % students in SJTU

Skills

Programming	Python, C/C++, $\LaTeX$, MATLAB, HTML, CSS, JavaScript
Frameworks	PyTorch, Tensorflow, NumPy, OpenCV, ROS, Flask
Simulators	IsaacGym, IsaacLab, ManiSkill (Sapien)
Robots	Unitree A1, Unitree Go2, Unitree G1, Franka Panda (+ self-built drones and robot arms)
Language	Chinese, English (TOEFL 112, GRE 328+3.0)